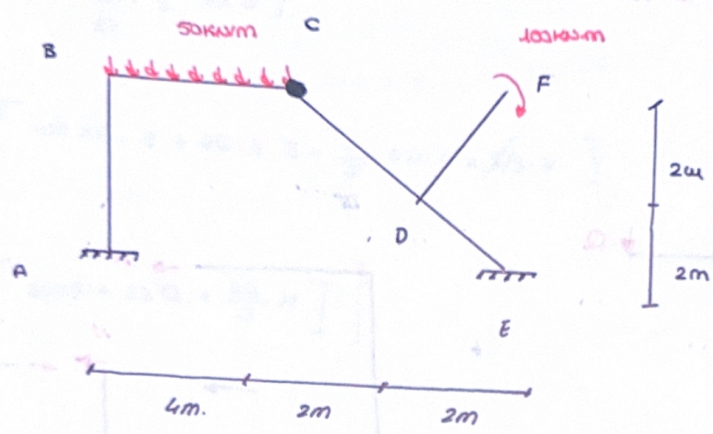


PROBLEMA CON PÓRTICA.

4. En la figura se muestra el análisis acotado de una estructura en la que cabe destacar que el nudo C es una rótula. Las cargas actuantes son: una sobrecarga vertical hacia abajo de 50 kN/m sobre la viga BC y un momento de $100 \text{ kN}\cdot\text{m}$ en el extremo F de la viga FE. El producto $EI = 6000 \text{ kN}\cdot\text{m}^2$ para todas las barras.

Se pide:

- Calcular las reacciones en los empotramientos.
- El giro en F.



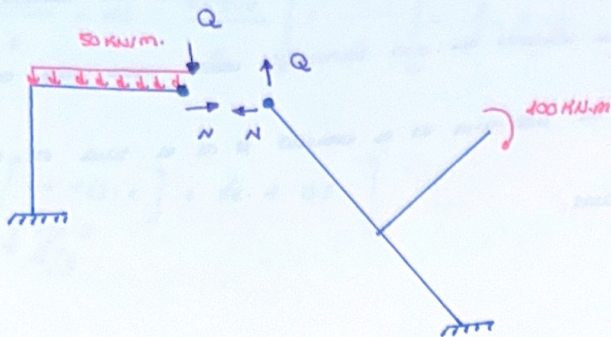
$6H = 2$ ya que la rótula me añade una ec. más de equilibrio

Mis ecuaciones de compatibilidad van a ser

$$\begin{aligned} \rightarrow u_{R1} &= u_{R2} \\ \downarrow v_{R1} &= \downarrow v_{R2} \end{aligned}$$

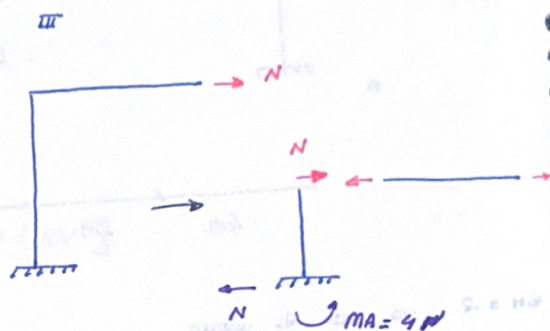
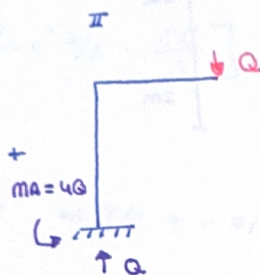
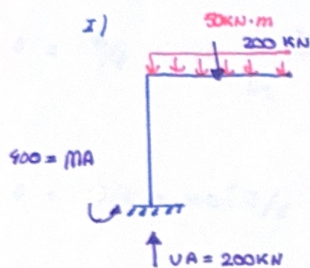
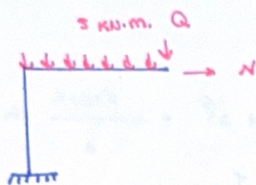
→ AMBAS CON EL MISMO SENTIDO.

Separamos por la estructura:



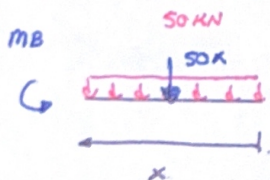
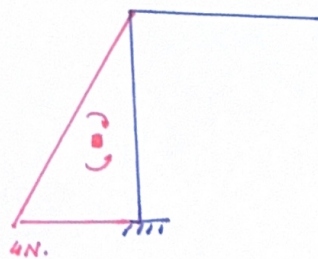
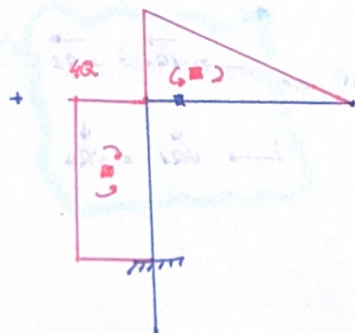
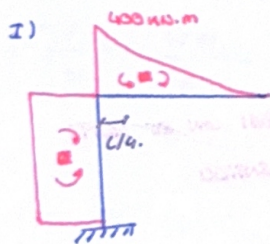
Buscamos conocer N y Q con las ec. de compatibilidad.

ESTADO I.



$$\circ E_{MA} = 0 \rightarrow MA - 200 \cdot 2 = 0 \rightarrow MA = 400 \text{ kN}\cdot\text{m}$$

Diagrama de Bending...



$$MB = 50x \cdot \frac{x}{2} \rightarrow MB = \frac{50x^2}{2}$$

$$MB(x=4) = \frac{50 \cdot 4^2}{2} = 400$$

$$V_C = \cancel{V_A} + \cancel{Q_A \cdot l} + \frac{1}{EI} [A \times d]$$

$$V_C = \frac{1}{EI} \left[400 \cdot 4 \times 4 + \frac{400 \cdot 4}{3} \times \frac{3}{4} \times 4 + 4Q \cdot 4 \cdot \frac{1}{4} + \frac{1}{2} \cdot 4Q \cdot 4 \times \frac{2}{3} \cdot 4 + \frac{1}{2} \cdot 4N \cdot 4 \times \frac{2}{3} \cdot 4 \right]$$

$$V_C = \frac{1}{EI} \left[6400 + 1600 + 64Q + \frac{64}{3}Q + 32N \right]$$

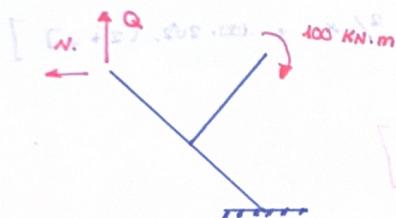
$$V_C = \frac{1}{EI} \left[8000 + \frac{256}{3}Q + 32N \right]$$

$$M_C = \cancel{M_A} + \cancel{Q_A \cdot l} + \frac{1}{EI} [A \times d]$$

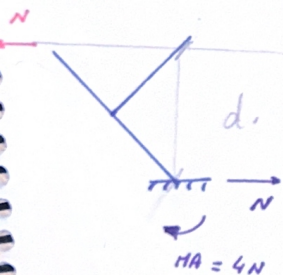
$$M_C = \frac{1}{EI} \left[400 \times 4 \cdot 2 + 4Q \cdot 4 \cdot 2 + \frac{1}{2} \cdot 4N \cdot 4 \times \frac{2}{3} \cdot 4 \right]$$

$$M_C = \frac{1}{EI} \left[3200 + 32Q + \frac{64}{3}N \right]$$

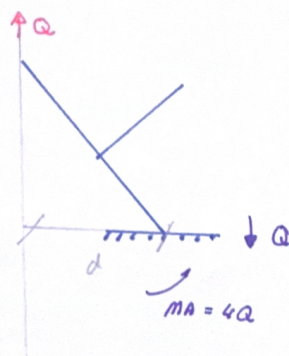
ESTADO II



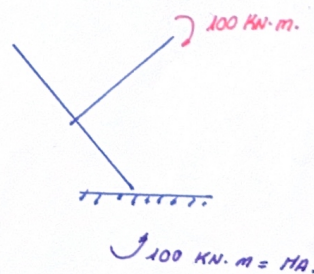
i)



ii)



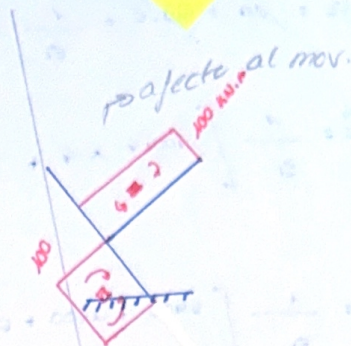
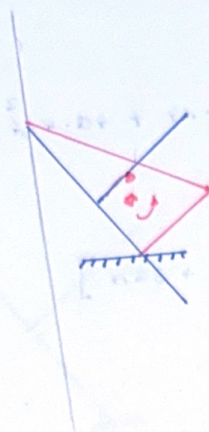
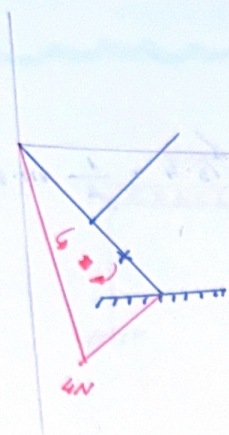
iii)



Área parábola: $\frac{q \cdot l^2}{3}$ flector

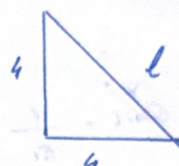
SI EL EJE DE MOVIMIENTO ESTÁ SOBRE LA BARRA NO AFECTA AL MOVIMIENTO.

Diagrama de flexiones.



$$\vec{v}_C = \vec{v}_E + \vec{\theta}_E \cdot l + \frac{1}{EI} [A \Delta d]$$

$$\vec{v}_C = \frac{1}{EI} \left[\frac{1}{2} \cdot 4N \cdot 4\sqrt{2} \cdot \frac{1}{3} \times 4 - \frac{1}{2} \cdot 4Q \cdot 4\sqrt{2} \cdot \frac{1}{3} \times 4 + 100 \cdot 2\sqrt{2} \cdot 1 \right]$$



$$l^2 = 4^2 + 4^2$$

$$l = \sqrt{4^2 + 4^2} = 4\sqrt{2}$$

$$\left[\vec{v}_C = \frac{1}{EI} \left[-\frac{64\sqrt{2}}{3} N - \frac{64\sqrt{2}}{3} Q + 28284 \right] \right]$$

$$\vec{u}_C = \vec{u}_E + \vec{\theta}_E \cdot l + \frac{1}{EI} [A \Delta d]$$

$$\vec{u}_C = \frac{1}{EI} \left[-\frac{1}{2} \times 4N \cdot 4\sqrt{2} \times \frac{1}{3} \times 4 + \frac{1}{2} \cdot 4Q \cdot 4\sqrt{2} \cdot \frac{2}{3} \times 4 + 100 \cdot 2\sqrt{2} \cdot (2 + 1) \right]$$

$$\left[\vec{u}_C = \frac{1}{EI} \left[-\frac{64\sqrt{2}}{3} N + \frac{64\sqrt{2}}{3} Q + 84853 \right] \right]$$

Iguando los movimientos:

$$\frac{1}{EI} \left[8000 + \frac{256}{3} Q + 32 N \right] = \frac{1}{EI} \left[\frac{64\sqrt{2}}{3} N - \frac{64\sqrt{2}}{3} Q + 282'84 \right]$$

$$\frac{1}{EI} \left[-\frac{64\sqrt{2}}{3} N + \frac{64\sqrt{2}}{3} Q + 848'53 \right] = \frac{1}{EI} \left[3200 + 32 Q + \frac{64}{3} N \right]$$

$$8000 + \frac{256}{3} Q + 32 N = \frac{64\sqrt{2}}{3} N - \frac{64\sqrt{2}}{3} Q + 282'84$$

$$7717'16 + 115'5 Q + 1'83 N = 0$$

$$-\frac{64\sqrt{2}}{3} N + \frac{64\sqrt{2}}{3} Q + 848'53 = 3200 + 32 Q + \frac{64}{3} N$$

$$\Rightarrow \begin{bmatrix} N = -43'3 \text{ KN} \\ Q = -66'13 \text{ KN} \end{bmatrix}$$

$$2351'47 + 1'83 Q + 5'5 N = 0$$